GRADE 7 ACCELERATED · UNIT 5 · LESSONS 8–9

Converting Mass, Area and Volume

Independent Set 5.08 **ANSWER KEY**

Name
Date
Period

Six questions, on your own, in silence. Show the step that does the work.

1. $1\,\text{lb} = 0.454\,\text{kg}$. How many kilograms is 30 lb?

13.62 kg

(30 ÷ 2.204 = 13.61 by the other ratio.)

2. $1\,\text{kg} = 2.204\,\text{lb}$. How many pounds is 25 kg?

55.1 lb

3. $1\,\text{gal} = 3.785\,\text{L}$. How many liters is 6 gal?

22.71 L

4. $1\,\text{ft} = 12\,\text{in}$. How many square inches is $9\,\text{ft}^2$?

1,296 in²

9 × 144: the ratio squared.

5. $1\,\text{c} = 8\,\text{fl oz}$ and $1\,\text{c} = 0.24\,\text{L}$. How many liters is 24 fl oz?

0.72 L

3 c × 0.24.

6. $1\,\text{ft} = 0.305\,\text{m}$. How many square meters is $30\,\text{ft}^2$? (Round to two places.)

About 2.79 m²

30 × 0.305²: the sheet's length ratio, squared, across systems.